DRAC Primer Session 2026

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Introduction

Welcome to the **Canadian Bioinformatics Workshops (CBW) DRAC Primer Session**. This hands-on course introduces a shared high-performance computing (HPC) training cluster and the commands needed to run reproducible bioinformatics analyses.

You will learn how to connect, manage files, transfer data, use software, submit and monitor Slurm jobs, and access interactive computing resources. The workshop finishes with a short bioinformatics challenge.

Slides

Please find below the slides for this session.

Workshop Details

Date: Wednesday, October 28, 2026. **Audience:** CBW participants who are new to Linux or HPC. **Platform:** DRAC-hosted CBW training cluster. The instructors will provide the registration link and workshop credentials; do not post passwords in shared documents.

Before the session

- Bring a laptop with a working internet connection and an SSH client (Terminal on macOS/Linux, or Windows Terminal/PowerShell with OpenSSH).
- Open the link `https://mokey.cbw.c3.ca`, and register your username, password and email address as instructed.
- This session targets the beginners who has little experience in command line, linux, and HPC.

- The login host for this workshop is `@cbw.c3.ca`; do not use a personal DRAC cluster account unless instructed.
- Use the workshop-specific instructions rather than assumptions about other DRAC clusters.

Course Outline

Date: Wednesday, October 28, 2026
Time: 11:00–15:00 PDT

Time (PDT)	Module / Activity
11:00–11:35	Module 1: HPC Infrastructure, registration and login
11:35–12:10	Module 2: Working with directories and files
12:10–12:20	**Break**
12:20–12:50	Module 3: Software environments and reproducibility
12:50–13:35	Module 4: Submit and monitor jobs with Slurm
13:35–13:45	**Break**
13:45–14:50	Module 5: Bioinformatics Mini-Challenge
14:50–15:00	Questions and Wrap-up

The session also includes instructor-led demonstrations, exercises and breaks. Follow the instructor's live timing announcements; the final timetable is distributed by CBW.

Getting Help

Ask an instructor during the session. You may share your screen if needed, but please avoid sharing passwords, or any other sensitive information.

Next: Module 1 — HPC infrastructure and access.

Chapter 1

Meet Your Faculty

1.0.1 Instructors

1.0.1.1 Jiarui Li

Research Support Analyst and Lead of Bioinformatics National Team (BNT) University of Alberta & Digital Research Alliance of Canada (DRAC) Edmonton, Alberta, Canada

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Jiarui (Jerry) Li is a Research Support Analyst at the University of Alberta and the Lead of the Bioinformatics National Team (BNT) at the Digital Research Alliance of Canada (DRAC). He received his bachelor's degree and Ph.D. in Genetics from Fudan University, China. Before joining the University of Alberta, he worked as a postdoctoral fellow and lecturer at Simon Fraser University, a lecturer at Shanghai University, Manager of the NGS Department at Applied Biological Materials Inc., and a Life Science Specialist at the University of British Columbia. He has broad experience in genetics, genomics, molecular biology, bioinformatics, and high-performance computing across both academic and industry settings.

1.0.2 Facilitator

1.0.2.1 Tian Rabbani

Regional Coordinator, CBH British Columbia Canadian Bioinformatics Hub, Simon Fraser University Vancouver, BC, Canada

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Tian completed her Master of Public Health in 2022 from Simon Fraser University and is currently a researcher and project lead at the Center for Infectious Disease Genomics and One Health (CIDGOH). She has been extensively involved in researching SARS-CoV-2 data-sharing through the Canadian COVID-19 Genomics Network (CanCOGeN), alongside leading efforts to understand metadata reporting in genomic epidemiology studies for outbreak and surveillance investigation. Her background in public health and knowledge in contextual data reporting intersect harmoniously to help inform current and future emergency response initiatives.

As British Columbia's regional coordinator for the Canadian Bioinformatics Hub, Tian is passionate about and dedicated to connecting bioinformatics communities and groups across the province. She aims to grow, develop, and expand data science and bioinformatics tools and training materials for prospective or new professionals, as well as for those looking to advance in their current careers. Tian believes accessibility, access, and availability of these tools and training opportunities are the cornerstone of growing our bioinformatics communities across the region.

Chapter 2

Module 1 - HPC infrastructure, registration and login

2.1 Learning goals

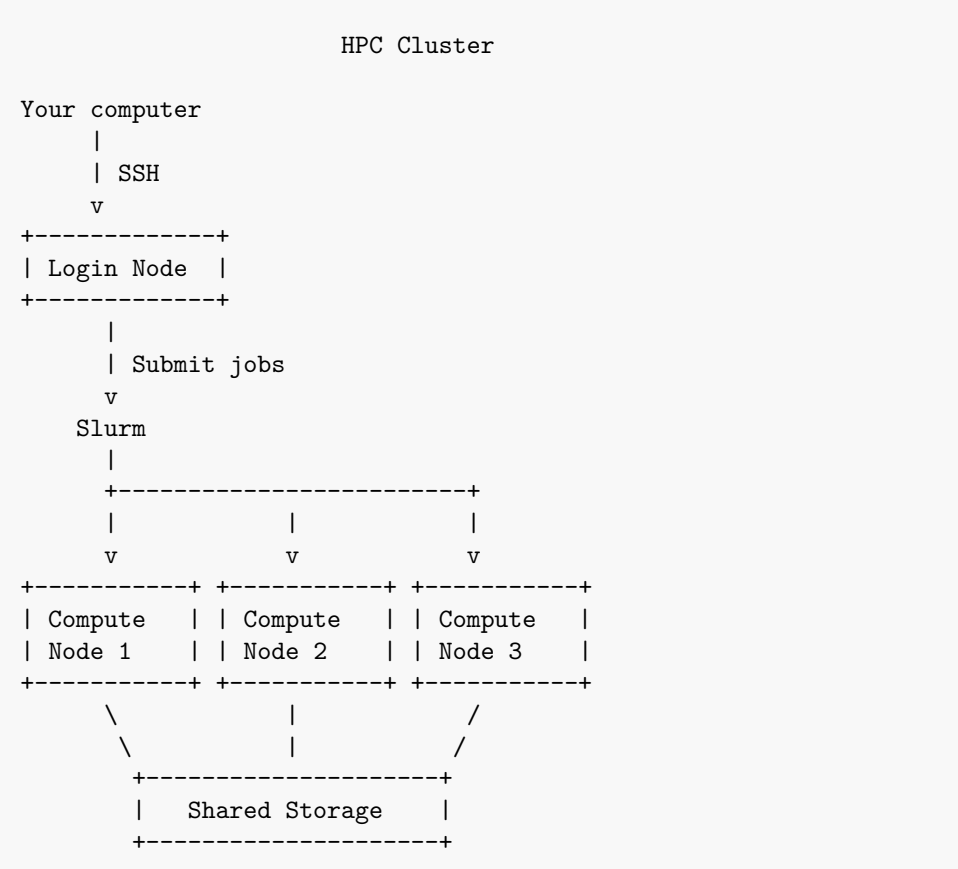
By the end of this module, you should be able to describe a training cluster, register for the workshop, and connect through SSH.

2.2 What is HPC?

High-performance computing (HPC) joins computing nodes, fast networking, shared storage, and a workload scheduler to support workloads that are too large or slow for a single laptop.

Component	Purpose
Login node	SSH access, lightweight editing, compilation, job submission
Compute node	CPU and memory resources for running analyses
Slurm	Queues, allocates and monitors jobs
Shared storage	Stores data accessible to jobs
Local node storage	Temporary space used during a job, when configured

2.3 Typical workflow



Important: Do not run computationally intensive analyses on login nodes. Submit batch jobs or request interactive allocations.

2.4 HPC terminology

- **CPU core:** unit of CPU computation requested by a job.
- **Memory (RAM):** memory required while a job runs.
- **Wall time:** maximum elapsed run time requested.
- **Job:** work the scheduler executes with an allocated resource request.
- **Partition:** scheduling queue or resource grouping (available partitions vary).

2.5 Register for the workshop

1. Open the `https://mokey.cbw.c3.ca` and click on **Create Account**.
2. Register the required **username, first name, last name, password and email**. The password must be at least **8 characters**.
3. Keep your credentials private. Use the workshop account, which may be distinct from a regular DRAC account.

2.6 Connect by SSH

Open Terminal (macOS/Linux) or Windows Terminal / PowerShell:

```
ssh YOUR_USERNAME@cbw.c3.ca
```

Replace `YOUR_USERNAME` with the username you registered. Confirm the host key only after verifying the hostname with the instructors, then enter your password when prompted. The password is not shown as you type.

Verify where you are:

```
whoami
hostname
pwd
```

To disconnect:

```
exit
```

2.7 Exercise: first login

1. Connect to `cbw.c3.ca`.
2. Run `whoami`, `hostname`, and `pwd`.
3. Identify which machine you are on and why intensive computation belongs on compute nodes.

Check your understanding

`whoami` prints your login name; `hostname` prints the machine name; `pwd` prints your current directory. Exact values depend on your account and the current training infrastructure.

2.8 Common connection issues

- **Permission denied:** Confirm the username and workshop password; do not share credentials.
- **Could not resolve hostname:** Check that `cbw.c3.ca` was entered correctly and that your network can resolve it.
- **Connection timed out:** Check connectivity or ask the instructor whether network access is restricted.

2.9 Backup solution:

- open `jupyter.cbw.c3.ca` in your browser
- login with your username and password
- Select **Terminal** in **User Interface**
- click on **logout** to exit.

Next: Module 2: Working with directories and files

Chapter 3

Module 2: Working with directories and files

3.1 Learning goals

Navigate Linux directories, inspect text files, organize results, and transfer files between your computer and the training cluster.

3.2 Explore the directory

```
pwd                    # Where am I?
ls                     # What is here?
ls -lh                 # Include modified dates, file sizes, etc.
mkdir -p cbw-primer    # Make a direcotory / make a folder
cd cbw-primer          # Navigate into the folder
cd ..                  # Go to the parent folder (move one level up)
pwd
cd ..                  # Now in /home
pwd
cd ..                  # Now in /
pwd
cd ..
pwd
ls -lh
cd project             # Now in /project
pwd
```

```
cd /home/USERNAME     # Replace USERNAME with your own username. Equivalent to: cd ~
pwd
cd ../../project      # Go back to /project with relative path
pwd
cd -                  # Go to the previous directory
pwd                   # /home/USERNAME
```

Important: Use _ to replace **space** in file or folder name.

Paths beginning with / are **absolute**; paths relative to your working directory are **relative**. .. refers to the parent directory and ~ refers to your home directory.

3.3 Create and inspect files

```
cd /home/USERNAME/cbw-primer
printf 'sample\treads\nA\t12\nB\t8\n' > counts.tsv
cat counts.tsv
head -n 2 counts.tsv
wc -l counts.tsv
cp counts.tsv counts-copy.tsv
mv counts-copy.tsv counts-backup.tsv
```

> creates or overwrites a file. Use it carefully. Use >> to append.

```
printf 'C\t15\n' >> counts.tsv
cat counts.tsv
```

You can use `nano counts.tsv` to edit the file. Cheat sheet: https://www.nano-editor.org/dist/latest/cheatsheet.html

3.4 File transfer from your own computer

Run the following commands in **a separate terminal on your laptop**, not inside the cluster SSH session:

```
# Download a result from the cluster
rsync -avh USERNAME@cbw.c3.ca:/home/USERNAME/cbw-primer/counts.tsv ./

# Upload one local file to your cluster home directory
rsync -avh ./counts.tsv USERNAME@cbw.c3.ca:/scratch/USERNAME
```

Change `USERNAME` and local paths as needed. `rsync` works over SSH and can resume incremental transfers. FileZilla is a graphical alternative: select **SFTP**, host `cbw.c3.ca`, port 22, and use your assigned workshop credentials.

3.5 Exercise

1. Create `/home/USERNAME/cbw-primer/results` and navigate into it.
2. Create a text file `samples.txt`, then inspect it with `cat` and `wc -l`.
3. Upload a small text file from your laptop using `rsync` or FileZilla.
4. Download the file back to your laptop.

Example commands

```
mkdir -p /home/USERNAME/cbw-primer/results
cd /home/USERNAME/cbw-primer/results
printf 'sample1\nsample2\nsample3\n' > ~/cbw-primer/samples.txt
cat samples.txt
wc -l ~/cbw-primer/samples.txt
```

From your laptop:

```
rsync -avh USERNAME@cbw.c3.ca:/home/USERNAME/cbw-primer/results/counts.tsv ./
```

3.6 Shared filesystem etiquette

Use descriptive filenames. Avoid generating millions of tiny files or copying large datasets unnecessarily. Ask instructors which directories are appropriate for persistent data and temporary job output; filesystem paths and quotas are cluster-specific.

Next: Module 3. Software environments and reproducibility.

Chapter 4

Module 3. Software environments and reproducibility

4.1 Learning goals

Find available software, load an environment module, and identify the commands and versions used in an analysis.

4.2 Environment modules

Clusters provide software through **environment modules**. Available applications and versions depend on the training system.

```
module avail
module list
module spider python       # Available on systems using Lmod
module load python         # If an appropriate Python module is available
python --version
module list
```

`module load` modifies your environment (for example `PATH`) for the current shell. When finished, you can use `module unload <module-name>` or `module reset`.

4.3 Python virtual environments

If Python and the `venv` module are available:

```
virtualenv --no-download ~/cbw-primer/venv
source ~/cbw-primer/venv/bin/activate
python --version
which python
pip install --no-index emoji
pip list
python -c 'import emoji;print(emoji.emojize(":elephant:"))'
deactivate
python -c 'import emoji;print(emoji.emojize(":elephant:"))'
```

4.4 For those who would like to use Conda:

Create a definition file, e.g. `alignment.def` and edit as needed:

```
Bootstrap: docker
From: ubuntu:22.04
Stage: build

%environment
 export LC_ALL=C
 source /usr/local/miniconda3/bin/activate alignment

%post
 apt update && apt install -y wget
 mkdir /conda
 cd /conda
 wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O minicon
 bash miniconda.sh -b -u -p /usr/local/miniconda3
 rm miniconda.sh
 export PATH="/usr/local/miniconda3/bin":$PATH
 eval "$(/usr/local/miniconda3/bin/conda shell.bash hook)"
 conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/main
 conda tos accept --override-channels --channel https://repo.anaconda.com/pkgs/r
 conda create -y -n alignment
 conda activate alignment
 conda install -n alignment -y bioconda::bwa
 rm -rf /conda
```

Create the container image, which is called singularity container image file (sif):

```
module load apptainer
apptainer build allignment.sif allignment.def
```

Run the software built in the container:

```
module load apptainer
apptainer exec alignment.sif bwa -h
```

4.5 Exercise

1. Find the latest version of SPAdes installed and load the module.
2. Find the installation path of loaded SPAdes.
3. Install numpy in venv.

Next: Module 4. Submit and monitor jobs with Slurm

Chapter 5

Module 4. Submit and monitor jobs with Slurm

5.1 Why use a scheduler?

On HPC, computation is submitted as a **job**. Slurm reserves cores, memory and time, and executes the job on an allocated compute node rather than on the login node.

5.2 Write a batch script

Create `hello.sbatch` with `nano hello.sbatch` and enter:

```
#!/bin/bash
#SBATCH --job-name=hello-hpc
#SBATCH --account=def-sponsor00
#SBATCH --time=00:05:00
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1000M

set -euo pipefail
printf 'Host: %s\n' "$(hostname)"
printf 'Job: %s\n' "$SLURM_JOB_ID"
printf 'CPUs: %s\n' "$SLURM_CPUS_PER_TASK"
```

```
sleep 10
printf 'Finished\n'
```

Submit and inspect:

```
sbatch hello.sbatch
sq
sacct --format=JobID,State,Elapsed,MaxRSS,ExitCode,NodeList
cat slurm_JOB_ID.out
```

Replace `JOB_ID` with the ID returned by `sbatch`. `sacct` reporting can lag and depends on the cluster's accounting configuration.

5.3 Understand resource requests

Slurm Option	Description	Analogy (Restaurant)
`--time`	Maximum wall-clock time for the job	Time allowed to prepare an order
`--ntasks`	Number of tasks/processes	Number of dishes being prepared simultaneously in an order
`--nodes`	Number of compute nodes	Number of kitchens
`--cpus-per-task`	Number of CPU cores per task	Number of chefs working on each dish
`--mem`	Memory allocated per node	Available workspace in each kitchen

Request resources based on measured usage instead of reserving much more than necessary. A job may remain **PENDING** while resources are unavailable.

5.4 Monitor and diagnose jobs

```
sq
sacct --format=JobID,JobName,State,Elapsed,MaxRSS,ReqTRES%60
scancel JOB_ID                  # Only when you intend to cancel the job
seff JOB_ID                      # Display the utilization
```

If you see `OUT_OF_MEMORY` or a time-limit failure, inspect logs and resource requests before rerunning.

5.5 Interactive allocations

5.5.1 salloc

For short tests and troubleshooting, request an interactive allocation:

```
salloc --time=00:15:00 --cpus-per-task=1 --mem=1G
srun --pty bash
hostname
htop                        # Press "q" to quit
echo $SLURM_TMPDIR          # Display the path to local temporary storage
exit                        # Exit the compute-node shell
exit                        # Release the allocation
```

The first `exit` closes the interactive shell on the compute node. The second `exit` leaves the allocation shell and releases the Slurm allocation.

5.5.2 JupyterHub

Open **the workshop JupyterHub** in your browser and follow the organizer's instructions.

A typical workflow is:

1. Sign in with your workshop credentials.
2. Select an available compute profile, if prompted.
3. Launch a notebook or terminal.
4. Shut down your session when finished.

5.6 Exercise

Submit a new job which runs the following command:

```
timeout 120s bash -c 'x=$(head -c 250M /dev/zero | tr "\0" x); sleep 120'
```

find its job ID, inspect its output log, and report whether the job completed successfully.

Next: Module 5. Bioinformatics Mini-Challenge.

Chapter 6

Module 5. Bioinformatics Mini-Challenge

6.1 Goal

Combine shell operations, a small sequence dataset, a reproducible script and a Slurm batch job. All data are generated inside your workspace; no external downloads are required.

6.2 1: Create a sample FASTA file

```
mkdir -p ~/cbw-primer/challenge
cd ~/cbw-primer/challenge
cat > sequences.fa <<'EOF'
>seqA
ACGTACGT
>seqB
GGGTTT
>seqC
ATATATATAT
EOF
```

FASTA records begin with > followed by an identifier and one or more sequence lines.

6.3 2: Explore the data

```
head sequences.fa
grep -c '^>' sequences.fa
wc -l sequences.fa
```

Question: How many sequences are present? Can I use `wc -l` then devided by 2 to count the number of sequences?

Answer

There are **3 sequences**. FASTA sequences can occupy multiple lines, so counting total lines does not count records reliably. Counting header lines beginning with `>` is appropriate for well-formed FASTA.

6.4 3: Build a simple analysis script

Save the following as `analyze.sh`:

```
#!/bin/bash
set -euo pipefail
input="${1:?Usage: bash analyze.sh INPUT.fa}"
printf 'Input: %s\n' "$input"
printf 'Sequences: '
grep -c '^>' "$input"
printf 'Total bases: '
awk '!/^>/ {gsub(/[[:space:]]/, ""); n += length($0)} END {print n+0}' "$input"
```

Run it:

```
bash analyze.sh sequences.fa
```

Expected results: **3 sequences** and **24 bases**. The base count sums sequence-line lengths and supports wrapped FASTA records.

6.5 4: Submit the analysis to Slurm

Save as `analyze.sbatch`:

```
#!/bin/bash
#SBATCH --account=def-sponsor00
#SBATCH --job-name=fasta-count
#SBATCH --time=00:05:00
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1000M

set -euo pipefail
cd "$SLURM_SUBMIT_DIR"
bash analyze.sh sequences.fa
```

Run:

```
sbatch analyze.sbatch
sq
```

Once it finishes, inspect the `slurm-JOBID.out` file (replace `JOBID` with the actual number).

6.6 5: Save and transfer results

From your laptop, retrieve the output with `rsync` (substitute the job ID and workshop username):

```
rsync -avh YOUR_USERNAME@cbw.c3.ca:~/cbw-primer/challenge/slurm-JOBID.out ./
```

6.7 6: Stretch challenge

Modify `analyze.sh` to accept multiple FASTA files, using a `for filename in "$@"` loop. Consider how you would process hundreds of files using a Slurm job array rather than running one long serial task.

6.8 7. Best practices

- Do NOT run the entire workflow in a single job!!!!!!

- Estimate the walltime, RAM, CPUs etc by benchmarking the tools
- Find the bottleneck of the tool
- Avoid duplicating the large public dataset such as NCBI BLAST database, Kraken2, Alphafold3, ColabFold, gnomAD ete, which might be found at: `/cvmfs/bio.data.computecanada.ca`
- Understand the pros/cons of different file systems, such as `$SLURM_TMPDIR`, `/scratch`, `/project`, `/home`, `/nearline`

6.9 Wrap-up

You have registered and logged in to an HPC training cluster, organized data, used environment modules, submitted and inspected a job, and run a reproducible bioinformatics script. For production work, benchmark resource usage and use the official cluster documentation and policies.